SQL Data Analysis Project

DecodeLabs Project 3
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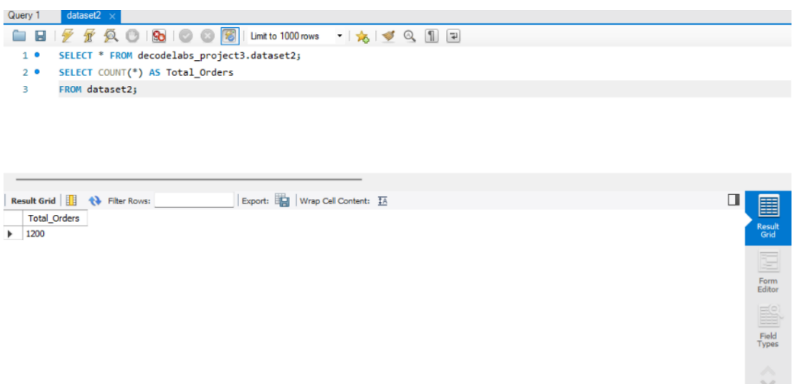
Objective : To analyze an e-commerce dataset using SQL and generate business insights through filtering, grouping, sorting, and aggregation.

Tools Used

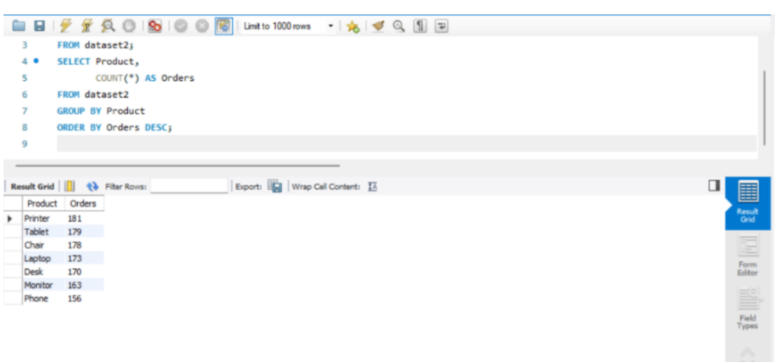
- MySQL Workbench
- SQL
- Excel

Query 1: Total Orders



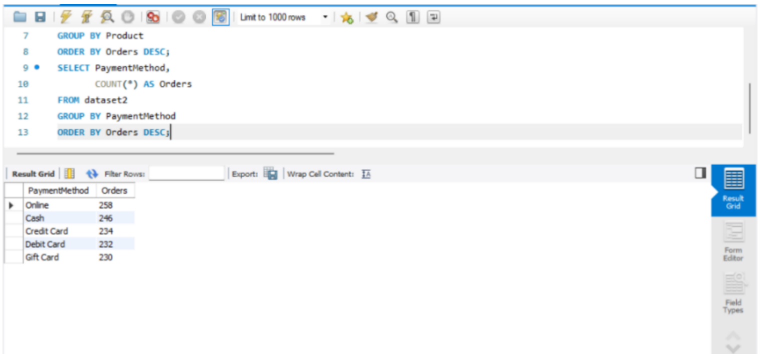
Purpose: Count the total number of orders in the dataset.

Query 2: Product Sales Analysis



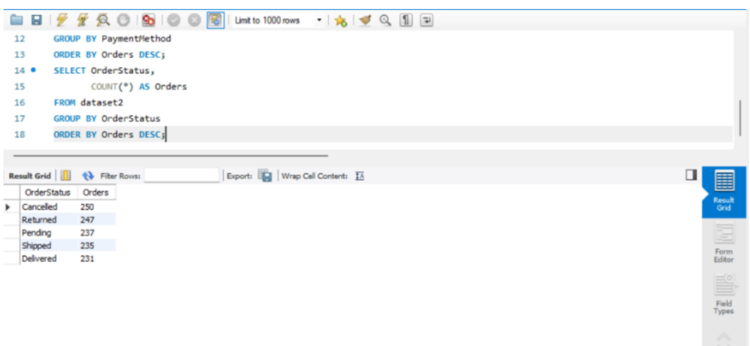
Purpose: Find the most frequently ordered products.

Query 3: Orders by Payment Method



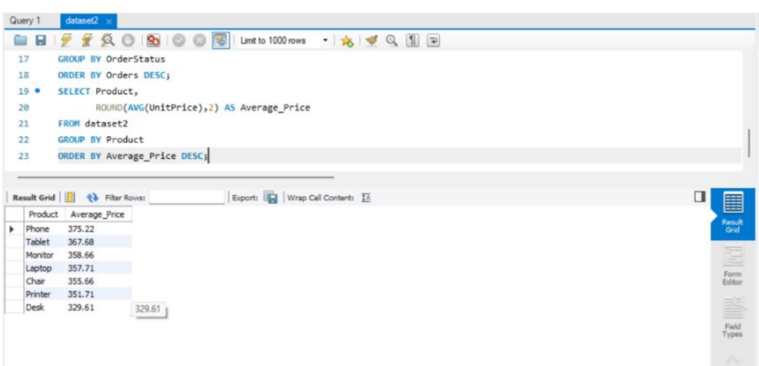
Purpose: Analyze customer payment preferences.

Query 4: Orders by Status



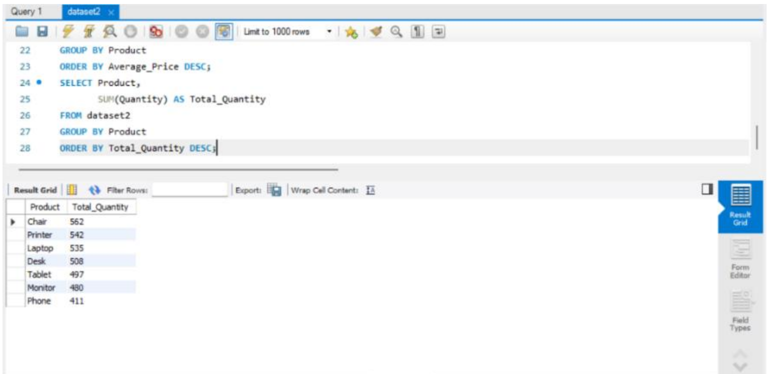
Purpose: View the distribution of order statuses.

Query 5: Average Product Price



Purpose: Compare average prices across products.

Query 6: Total Quantity Sold by Product

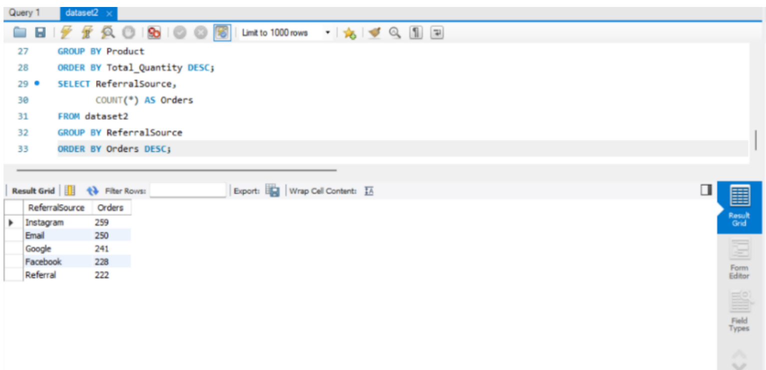


```sql
GROUP BY Product
ORDER BY Average_Price DESC;
SELECT Product,
       SUM(Quantity) AS Total_Quantity
FROM dataset2
GROUP BY Product
ORDER BY Total_Quantity DESC;
```

Product	Total_Quantity
Chair	562
Printer	542
Laptop	535
Desk	508
Tablet	497
Monitor	480
Phone	411

Purpose: Identify the products with the highest sales volume.

Query 7: Referral Source Analysis

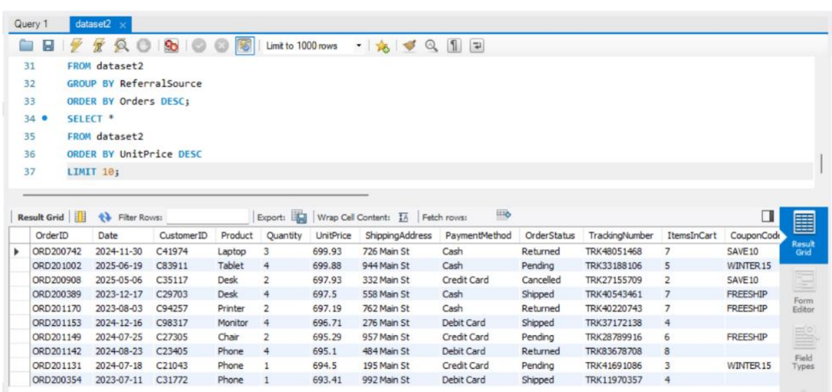


```sql
GROUP BY Product
ORDER BY Total_Quantity DESC;
SELECT ReferralSource,
       COUNT(*) AS Orders
FROM dataset2
GROUP BY ReferralSource
ORDER BY Orders DESC;
```

ReferralSource	Orders
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Purpose: Determine which marketing/referral channels generate the most orders.

Query 8: Top 10 Most Expensive Orders



```sql
FROM dataset2
GROUP BY ReferralSource
ORDER BY Orders DESC;
SELECT *
FROM dataset2
ORDER BY UnitPrice DESC
LIMIT 10;
```

OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode
ORD200742	2024-11-30	C41974	Laptop	3	699.93	726 Main St	Cash	Returned	TRK48051468	7	SAVE10
ORD201002	2025-06-19	C83911	Tablet	4	699.88	944 Main St	Cash	Pending	TRK33188106	5	WINTER15
ORD200908	2025-05-06	C35117	Desk	2	697.93	332 Main St	Credit Card	Cancelled	TRK27155709	2	SAVE10
ORD200389	2023-12-17	C29703	Desk	4	697.5	558 Main St	Cash	Shipped	TRK40543461	7	FREESHIP
ORD201170	2023-08-03	C94257	Printer	2	697.19	762 Main St	Cash	Returned	TRK40220743	7	FREESHIP
ORD201153	2024-12-16	C98317	Monitor	4	696.71	276 Main St	Debit Card	Shipped	TRK37172138	4	
ORD201149	2024-07-25	C27305	Chair	2	695.29	957 Main St	Credit Card	Pending	TRK28789916	6	FREESHIP
ORD201142	2024-08-23	C23405	Phone	4	695.1	484 Main St	Debit Card	Returned	TRK83678708	8	
ORD201131	2024-07-18	C21043	Phone	1	694.5	195 Main St	Credit Card	Pending	TRK41691086	3	WINTER15
ORD200354	2023-07-11	C31772	Phone	1	693.41	992 Main St	Debit Card	Shipped	TRK11970357	4	

Purpose: Display the highest-priced orders in the dataset.

Conclusion : This project demonstrated the use of SQL for data extraction, filtering, grouping, and aggregation. Through SQL queries, meaningful business insights were generated from an e-commerce dataset.